CONSERVED CHARGES IN LINEARLY REINFORCED GRAPH URNS

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Abstract. In the graph-based Pólya urn with linear reinforcement (WARM with $\alpha = 1$) in continuous time, every vector σ in the kernel of the unsigned vertex–edge incidence matrix yields an exact martingale $M_\sigma = \sum_e \sigma_e H(N_e - 1)$, where H is the harmonic number. On every finite graph its terminal value has the law of $\sum_e \sigma_e \log W_e$ with independent standard exponential variables W_e. On the even cycle C_L the normalised counts converge to $(1+u, 1-u, \dots)$ with $L\,\mathrm{artanh}(u)$ equal to the terminal value of the charge, so that $u = \tanh(M/L)$ where M is a sum of $L/2$ independent standard logistic variables, and $\sqrt{L}\,\mathrm{artanh}(u)$ converges in law to the centred normal law with variance $\pi^2/6$. This gives the law of the random limit for the continuous-time, one-vertex-at-a-time analogue of the urn studied by Chen and Lucas and by Lima. On the integer line the expected quadratic variation of the charge per vertex is $\pi^2/6$.

1. The model and the result

In the urn with graph-based competition of van der Hofstad, Holmes, Kuznetsov and Ruszel [4] (WARM), each edge of a graph carries a count, each vertex carries a Poisson clock, and when a vertex rings it reinforces one of its edges with probability proportional to N_e^α. The discrete-time urn, in which every edge adds a ball to one of its two endpoints at each step, was introduced by Benaïm, Benjamini, Chen and Lima [2]. At $\alpha = 1$ it was studied by Chen and Lucas [3], who proved convergence on regular bipartite graphs to a random point of a segment of equilibria and left the law of that point open, and by Lima [5], who completed the convergence proof for balanced bipartite graphs and asked for the distribution of the limit (his Question 1). On a cycle the two models are related by exchanging vertices and edges: the bins of the discrete-time urn on C_L correspond to the edges of WARM on C_L, and its competing pairs to the vertices of WARM. The discrete-time urn updates every pair at each step, whereas WARM updates one vertex at a time at Poisson times. The results below concern WARM; no transfer of the limit law to the synchronous urn is proved. On the infinite

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line, Kleptsyn conjectures that the normalised counts of the critical process converge to 1 [1, Section 5].

The random limit is fixed by conserved charges. For every σ in the kernel of the unsigned incidence matrix, $M_\sigma = \sum_e \sigma_e H(N_e - 1)$ is an exact martingale whose quadratic variation never exceeds $\zeta(2)\sum_e \sigma_e^2$ (Theorem 1); up to a vanishing error it equals $\sum_e \sigma_e \log(N_e/t)$, which is conserved by the mean-field flow. Its terminal value has an explicit law on every finite graph: $\mathbb{E}e^{i\theta M_{\sigma,\infty}} = \prod_e \Gamma(1 + i\theta\sigma_e)$, so that the charges jointly have the law of $(\sum_e \sigma_e \log W_e)_\sigma$ with independent standard exponential W_e (Theorem 3). On the even cycle C_L the kernel is one-dimensional, the normalised counts converge to $(1+u_\infty, 1-u_\infty, \dots)$ with $L\,\mathrm{artanh}(u_\infty) = M_\infty$ (Theorem 6), and therefore $u_\infty = \tanh(M_\infty/L)$ with M_∞ a sum of $L/2$ independent standard logistic variables (Corollary 7). In particular $\sqrt{L}\,\mathrm{artanh}(u_\infty) \Rightarrow \mathcal{N}(0, \pi^2/6)$, and $\pi^2/6$ is the expected quadratic variation of the charge per vertex, on every even cycle and on $\mathbb{Z}$ (Proposition 8). Section 5 reports simulations: the terminal law on cycles, random limits on C_4 and K_4 whose dimension matches the kernel, and the slow decay of the alternating envelope on long cycles that the charges predict for the line.

2. The charge martingales

Let $G = (V, E)$ be a finite graph without loops or isolated vertices, with maximum degree d. Each vertex carries an independent rate-one Poisson clock; when the clock of v rings, one edge $e \ni v$ is chosen with probability N_e/S_v, where $S_v = \sum_{f \ni v} N_f$, and N_e increases by one. All counts start at 1. Constants C, C', C'', c are finite, random where stated, and may differ between proofs. Write $X_e(t) = N_e(t)/t$, let B be the unsigned incidence matrix, $B_{ve} = 1$ when $v \in e$, and let $H^{(p)}(n) = \sum_{k=1}^{n} k^{-p}$, $H = H^{(1)}$.

Theorem 1. *Let $\sigma \in \mathbb{R}^E$ satisfy $\sum_{e \ni v} \sigma_e = 0$ for every v. Then*

$$M_\sigma(t) = \sum_{e \in E} \sigma_e\, H(N_e(t) - 1)$$

is a martingale with $M_\sigma(0) = 0$, jumps bounded by $\max_e |\sigma_e|$, predictable quadratic variation

$$\langle M_\sigma \rangle_t = \int_0^t \sum_{v \in V} \frac{1}{S_v(s)} \sum_{e \ni v} \frac{\sigma_e^2}{N_e(s)}\, ds,$$

and quadratic variation $[M_\sigma]_t = \sum_e \sigma_e^2 H^{(2)}(N_e(t) - 1) \le \zeta(2)\sum_e \sigma_e^2$. In particular M_σ is bounded in L^2.

Proof. When N_e moves from n to $n+1$, $H(N_e - 1)$ increases by $1/n$. A ring at v therefore changes M_σ by σ_e/N_e with probability N_e/S_v, for each $e \ni v$. The conditional mean of the change is $S_v^{-1}\sum_{e \ni v} \sigma_e = 0$, and its conditional second moment is $S_v^{-1}\sum_{e \ni v} \sigma_e^2/N_e$. On $[0, T]$ the number of rings is Poisson with mean $|V|T$ and every jump is at most $\max_e |\sigma_e|$, so

the compensated sums are square-integrable martingales. The jump at the passage of N_e from n to $n+1$ is σ_e/n, and each value n is passed at most once, which gives the formula and the bound for $[M_\sigma]_t$; hence $\mathbb{E}M_\sigma(t)^2 = \mathbb{E}[M_\sigma]_t \le \zeta(2)\sum_e \sigma_e^2$. □

For $\sigma \notin \ker B$ the same computation gives the drift $\sum_v (B\sigma)_v/S_v$, which is not zero in general. The space $\ker B$ has dimension $|E| - |V| + \beta$, with β the number of bipartite components. It is nonzero exactly when some component contains an even cycle or two odd cycles. Summing the vertex constraints gives $\sum_e \sigma_e = 0$ for $\sigma \in \ker B$, and $H(n-1) = \psi(n) + \gamma$, where ψ is the digamma function and γ is Euler's constant, so

$$\sum_e \sigma_e \log X_e(t) = M_\sigma(t) + \sum_e \sigma_e\big(\log N_e(t) - \psi(N_e(t))\big), \tag{1}$$

with $\log n - \psi(n) = \frac{1}{2n} + O(n^{-2})$. The charges $\sum_e \sigma_e \log X_e$ are therefore martingales up to a term that vanishes as every count grows. In the mean-field equation $\dot{x}_e = x_e \sum_{v\in e} S_v(x)^{-1} - x_e$ (time $\log t$) they are exactly conserved: $\frac{d}{d\tau}\sum_e \sigma_e \log x_e = \sum_v S_v(x)^{-1}\sum_{e\ni v}\sigma_e - \sum_e \sigma_e = 0$.

Lemma 2. *For an edge $e = (u, v)$ let $a_e(t) = \int_0^t \big(S_u(s)^{-1} + S_v(s)^{-1}\big)ds$ be its internal time. Then $D_e(t) = H(N_e(t)-1) - a_e(t)$ converges almost surely and in L^2. Almost surely $\liminf_{t\to\infty} S_v(t)/t \ge 1$ for every v, and for every $\lambda < 2/(d+1)$ and every edge e, $\liminf_{t\to\infty} N_e(t)/t^\lambda > 0$; in particular every count tends to infinity.*

Proof. The count N_e increases at rate $N_e(S_u^{-1} + S_v^{-1}) \le 2$, and each increase from n to $n+1$ adds $1/n$ to $H(N_e - 1)$, so D_e is a martingale. Its quadratic variation is $H^{(2)}(N_e(t) - 1) \le \zeta(2)$, so it is bounded in L^2 and converges almost surely and in L^2. Let $F_w(t)$ be the number of rings of w by time t. Every ring of v adds one to S_v, so $S_v \ge F_v$ and $\liminf S_v/t \ge 1$. A ring of w adds to S_v only if $w = v$ or $w \sim v$, and then by one, so $S_v(t) \le d + \sum_{w\in\{v\}\cup N(v)} F_w(t)$. Since $F_w(t)/t \to 1$ almost surely, for $\varepsilon > 0$ there is an almost surely finite T with $S_w(t) \le (d+1)(1+\varepsilon)t$ for all w and $t \ge T$. Then $a_e(t) \ge \lambda' \log t - C$ for $t \ge T$, with $\lambda' = 2/((d+1)(1+\varepsilon))$ and C finite. Since D_e converges, $H(N_e(t) - 1) \ge \lambda' \log t - C'$, so $N_e \to \infty$ and, as $H(n-1) - \log n \to \gamma$, $\log N_e(t) \ge \lambda' \log t - C''$. Taking ε small gives the claim. □

For $\sigma \in \ker B$ the internal times cancel,

$$\sum_e \sigma_e a_e(t) = \int_0^t \sum_v \frac{(B\sigma)_v}{S_v(s)}\, ds = 0,$$

so that $M_\sigma = \sum_e \sigma_e D_e$.

Theorem 3. *Let $\sigma \in \ker B$. Then $M_\sigma(t)$ converges almost surely and in L^2 to a limit $M_{\sigma,\infty}$, and*

$$\mathbb{E}\, e^{i\theta M_{\sigma,\infty}} = \prod_{e\in E} \Gamma(1 + i\theta\sigma_e), \qquad \theta \in \mathbb{R}.$$

Equivalently, $M_{\sigma,\infty}$ has the law of $\sum_e \sigma_e \log W_e$, where the W_e are independent standard exponential variables; since M_σ is linear in σ, the vector $(M_{\sigma,\infty})_{\sigma\in\ker B}$ has the law of $(\sum_e \sigma_e \log W_e)_{\sigma\in\ker B}$. Moreover $\mathbb{E}M_{\sigma,\infty}^2 = \mathbb{E}\langle M_\sigma\rangle_\infty = \zeta(2)\sum_e \sigma_e^2$ and $\operatorname{Var}\langle M_\sigma\rangle_\infty = \zeta(4)\sum_e \sigma_e^4$.

Proof. Convergence follows from Theorem 1. For real θ and positive integers n_e put

$$P_\theta(n) = \prod_e \frac{\Gamma(n_e + i\theta\sigma_e)}{\Gamma(n_e)\,\Gamma(1 + i\theta\sigma_e)}.$$

An increase of n_e multiplies P_θ by $1 + i\theta\sigma_e/n_e$, so the generator of the urn applied to P_θ is $P_\theta(n)\sum_v S_v^{-1}\sum_{e\ni v} n_e \cdot i\theta\sigma_e/n_e = i\theta P_\theta(n)\sum_v (B\sigma)_v/S_v = 0$. Since $|\Gamma(x+iy)| \le \Gamma(x)$ for $x > 0$, $|P_\theta| \le \prod_e |\Gamma(1+i\theta\sigma_e)|^{-1}$. A bounded function with zero generator of a jump process whose total jump rate is $|V|$ is a martingale, so $\mathbb{E}P_\theta(N(t)) = 1$. As $n \to \infty$, $\Gamma(n+z)/(\Gamma(n)n^z) \to 1$ for fixed complex z, and every count tends to infinity (Lemma 2), so $P_\theta(N(t))\prod_e \Gamma(1 + i\theta\sigma_e) - \exp(i\theta\sum_e \sigma_e \log N_e(t)) \to 0$. Since $\sum_e \sigma_e = 0$, (1) gives $\sum_e \sigma_e \log N_e(t) - M_\sigma(t) \to 0$. By dominated convergence, $\mathbb{E}e^{i\theta M_{\sigma,\infty}}\prod_e \Gamma(1 + i\theta\sigma_e)^{-1} = 1$. For a standard exponential W, $\mathbb{E}W^{i\theta} = \Gamma(1 + i\theta)$, which gives the second form, and the joint law follows by the Cramér–Wold device.

Since every count tends to infinity, $[M_\sigma]_\infty = \zeta(2)\sum_e \sigma_e^2$, and $\mathbb{E}M_{\sigma,\infty}^2 = \mathbb{E}[M_\sigma]_\infty = \mathbb{E}\langle M_\sigma\rangle_\infty$ by monotone convergence. The process $R = [M_\sigma] - \langle M_\sigma\rangle$ is a martingale whose jumps are the squared jumps of M_σ, and $\langle M_\sigma\rangle$ is continuous, so $[R]_t = \sum_e \sigma_e^4 H^{(4)}(N_e(t) - 1) \le \zeta(4)\sum_e \sigma_e^4$. Hence R is bounded in L^2, and as $[M_\sigma]_\infty$ is deterministic, $\operatorname{Var}\langle M_\sigma\rangle_\infty = \mathbb{E}R_\infty^2 = \mathbb{E}[R]_\infty = \zeta(4)\sum_e \sigma_e^4$. □

The law in Theorem 3 is the one obtained by letting each count grow as an independent Yule process run on its internal time a_e: such a process satisfies $\log N_e - a_e \to \log W_e$, and the internal times cancel in every charge.

Remark 4. Suppose that $X(t)$ converges to a point x_∞ with positive coordinates and deterministic vertex sums $Bx_\infty = b$. By (1), $\sum_e \sigma_e \log x_{\infty,e} = M_{\sigma,\infty}$ for every $\sigma \in \ker B$. These values determine x_∞: on $P = \{x \in (0,\infty)^E : Bx = b\}$, assumed nonempty, the map $x \mapsto \pi(\log x)$, with π the orthogonal projection onto $\ker B$, is a bijection onto $\ker B$. Indeed, for $m \in \ker B$ the strictly convex function $\sum_e x_e(\log x_e - 1) - \langle m, x\rangle$ has a unique minimiser on the compact closure of P; it lies in P, because moving from a boundary point towards a point of P decreases the function at infinite rate, and the first-order condition there is $\pi(\log x) = m$; injectivity follows from $\sum_e (x_e - y_e)(\log x_e - \log y_e) > 0$ for distinct $x, y \in P$. Since $(\ker B)^\perp$ is the

range of $B^\top$, Theorem 3 then says that x_∞ has the law of the unique point of P of the form $x_e = W_e\, e^{y_u+y_v}$, $e=(u,v)$, with vertex potentials y: the scaling of independent exponential edge weights to the vertex sums b.

3. Convergence on the even cycle

On the even cycle C_L (edges indexed mod L, vertex v between edges $v-1$ and v) the kernel is spanned by $\sigma_e=(-1)^e$, and since $S_v = N_{v-1}+N_v$,

$$\frac{1}{S_v}\Big(\frac{1}{N_{v-1}}+\frac{1}{N_v}\Big)=\frac{1}{N_{v-1}N_v}, \qquad \langle M\rangle_t=\int_0^t \sum_v \frac{ds}{N_{v-1}(s)N_v(s)},$$

where $M=M_\sigma$. Let $L\ge 4$ be even. With K_e the compensated count of $e=(u,v)$, and since $S_w(N(t))=tS_w(X(t))$,

$$dX_e(t)=\frac{F_e(X(t))}{t}\,dt+\frac{dK_e(t)}{t}, \qquad F_e(x)=x_e\,\partial_e\mathcal{L}(x),$$

so in the time $\tau=\log t$ the normalised counts follow, up to the martingale term, the field

$$\frac{dx_e}{d\tau}=x_e\,\partial_e\mathcal{L}(x), \qquad \mathcal{L}(x)=\sum_v \log S_v(x)-\sum_e x_e, \tag{2}$$

with $\partial_e\mathcal{L}=S_u^{-1}+S_v^{-1}-1$ for $e=(u,v)$. A direct computation [4] gives

$$\frac{d}{d\tau}\mathcal{L}=\sum_e x_e(\partial_e\mathcal{L})^2\ge 0, \qquad \frac{d}{d\tau}\sum_e x_e = L-\sum_e x_e.$$

Lemma 5. *The equilibria of* (2) *in* $\{x\in[0,\infty)^E : \sum_e x_e = L,\ S_v(x)>0\ \forall v\}$ *are*

1. *the segment* $\Sigma=\{x^{(u)} : u\in[-1,1]\}$, $x_e^{(u)}=1+(-1)^e u$, *on which* $S_v\equiv 2$;
2. *finitely many points, each with an edge* e *such that* $x_e=0$ *and* $\partial_e\mathcal{L}(x)\ge 1/(2L)$.

Proof. Let x be an equilibrium. A vertex with both edges at zero has $S_v=0$, which is excluded. If every edge is positive, $1/S_u+1/S_v=1$ along every edge of the cycle, so S takes two values s,s' alternately; summing, $\sum_v S_v = 2\sum_e x_e = 2L$ gives $s+s'=4$, hence $s=s'=2$ and $x\in\Sigma$. Otherwise the positive edges form paths $w_0w_1\cdots w_k$ separated by single zero edges, and along such a path $S(w_i)$ alternates between s and $s'=s/(s-1)$. Since the edges outside the path at w_0 and w_k are zero, $x_1=S(w_0)=s$, and solving along the path gives $x_{2i}=id$, $x_{2i+1}=s-id$ with $d=s'-s$. If k is odd the end condition $x_k=S(w_k)=s'$ forces $d=0$, hence $k=1$ and $x_1=2$. If k is even it forces $d=2s/k$, and $1/s+1/s'=1$ gives $s=(2k+2)/(k+2)$; all x_j are then positive. So every block is either a single edge of weight 2 with $S=2$ at both ends, or an even block of length k with $S=(2k+2)/(k+2)$ at both ends, and the configuration is determined by the block lengths. If all blocks have length one, x alternates $2,0$ and lies in Σ ($u=\pm1$). Otherwise

some even block of length $k \le L-1$ meets a zero edge e whose other end has $S \in \{2, (2k'+2)/(k'+2)\}$, so $\partial_e \mathcal{L} \ge \frac{k+2}{2k+2} + \frac{1}{2} - 1 = \frac{1}{2k+2} \ge \frac{1}{2L}$, the smallest value occurring when the other end has $S = 2$. □

Theorem 6. *On C_L, L even, almost surely $X(t) \to x^{(u_\infty)}$ with $|u_\infty| < 1$ and*

$$L \operatorname{artanh}(u_\infty) = M_\infty := \lim_{t\to\infty} \sum_e (-1)^e H(N_e(t) - 1).$$

Proof. By Lemma 2 and since $\sum_e N_e(t) - L$ is a Poisson process of rate L, $X(t)$ eventually stays in the compact set $K = \{x \in [0,\infty)^E : \sum x_e \le 2L,\ S_v(x) \ge 1/2\ \forall v\}$, which is forward invariant for (2) and on which the field is Lipschitz. The total jump rate is L, so by Doob's inequality the martingale term over a window satisfies $\mathbb{E} \sup_{h\le T} |\int_{e^\tau}^{e^{\tau+h}} t^{-1} dK_e(t)|^2 \le 4\int_{e^\tau}^{e^{\tau+T}} Lt^{-2}\,dt \le 4Le^{-\tau}$, and the Borel–Cantelli lemma along integer τ shows that it tends to zero almost surely, uniformly over windows of length T. The jumps of X are at most $1/t$. By Gronwall's lemma the linear interpolation of $\tau \mapsto X(e^\tau)$ is then a precompact asymptotic pseudotrajectory of the flow of (2) on K in the sense of [6, Section 3], and its limit set is internally chain transitive [6, Theorem 5.7]. The function $-\mathcal{L}$ is strictly decreasing along every nonstationary orbit, and by Lemma 5 it takes finitely many values on the equilibria ($\mathcal{L} \equiv L\log 2 - L$ on Σ). By [6, Proposition 6.4] the limit set consists of equilibria, and being connected it lies in Σ or is a single point of type (2).

Type (2): let x^* be such a point, with $x^*_e = 0$ and $\partial_e \mathcal{L}(x^*) \ge 1/(2L)$, and choose a neighbourhood U of x^* in K on which $x_e \le 1$ and $\partial_e \mathcal{L} \ge 1/(4L)$. On the event $A_m = \{X(t) \in U \text{ for all } t \ge m\}$, $m \in \mathbb{N}$, the internal time of e satisfies $a_e(t) - a_e(m) = \int_m^t (1 + \partial_e \mathcal{L}(X(s)))\, s^{-1} ds \ge (1+\delta)\log(t/m)$ with $\delta = 1/(4L)$. By Lemma 2, $\log N_e(t) - a_e(t)$ converges, so $X_e(t) \ge ct^\delta \to \infty$ on A_m, which contradicts $x_e \le 1$ on U. Hence $\mathbb{P}(A_m) = 0$ for every m, and X does not converge to x^*. As there are finitely many such points, the limit set lies in Σ.

By (1) and Theorem 1, $\Phi(X(t)) := \sum_e (-1)^e \log X_e(t) \to M_\infty$, finite almost surely. Along a sequence of times with $X \to x^{(1)}$ every even coordinate tends to 2 and every odd coordinate to 0, so $\Phi \to +\infty$; similarly $\Phi \to -\infty$ near $x^{(-1)}$. So neither endpoint is a limit point, and the limit set is a compact subset of $\{x^{(u)} : |u| < 1\}$, where Φ is continuous and $\Phi(x^{(u)}) = \frac{L}{2}\log\frac{1+u}{1-u} = L\operatorname{artanh}(u)$ is strictly increasing in u. Every limit point $x^{(u)}$ therefore has $L\operatorname{artanh}(u) = M_\infty$, and the limit set is the single point $x^{(u_\infty)}$. □

Corollary 7. *On C_L, $L \ge 4$ even, $u_\infty = \tanh(M_\infty/L)$, where M_∞ has the law of $\Lambda_1 + \cdots + \Lambda_{L/2}$ with independent standard logistic variables Λ_j (density $e^{-s}/(1+e^{-s})^2$), that is,*

$$\mathbb{E}\, e^{i\theta M_\infty} = \Big(\frac{\pi\theta}{\sinh \pi\theta}\Big)^{L/2}.$$

In particular $\mathbb{E}M_\infty^2 = L\pi^2/6$, $\mathbb{E}\big(\langle M\rangle_\infty/L - \pi^2/6\big)^2 = \pi^4/(90L)$, *and as* $L \to \infty$ *through even values, both* $\sqrt{L}\,\mathrm{artanh}(u_\infty)$ *and* $\sqrt{L}\,u_\infty$ *converge in law to* $\mathcal{N}(0, \pi^2/6)$.

Proof. By Theorem 3 with $\sigma_e = (-1)^e$, M_∞ has the law of $\sum_j(\log W_{2j} - \log W_{2j+1})$, and $\mathbb{E}e^{i\theta(\log W - \log W')} = \Gamma(1+i\theta)\Gamma(1-i\theta) = \pi\theta/\sinh(\pi\theta)$ for independent standard exponential W, W', the characteristic function of the standard logistic law. Theorem 6 gives $u_\infty = \tanh(M_\infty/L)$, and Theorem 3 gives the two moments. The logistic law has variance $\pi^2/3$, so the central limit theorem gives $M_\infty/\sqrt{L} \Rightarrow \mathcal{N}(0, \pi^2/6)$, and $\sqrt{L}\,u_\infty - M_\infty/\sqrt{L} \to 0$ in probability since $\tanh(x)/x \to 1$ as $x \to 0$ and $M_\infty/L \to 0$. □

On C_4, with $p = (1+u_\infty)/2$, the density of M_∞ is $f(s) = (s\coth(s/2) - 2)/(4\sinh^2(s/2))$, the density of p is $2f(2\log\frac{p}{1-p})/(p(1-p))$, which behaves like $4(1-p)\log\frac{1}{1-p}$ as $p \to 1$, and $\mathrm{Var}\,p = \pi^2/32 - 1/4 = 0.05843\ldots$. The limit law on C_4 is therefore not a beta law.

4. The infinite line

On $\mathbb{Z}$, with vertex v between edges $v-1$ and v, the process is built from the same rate-one clocks and choices. A vertex w that does not ring during $[0,T]$ separates the line on that interval: the edges to the left of w change only at rings to the left of w, which depend only on those edges, and similarly on the right. Each vertex is silent on $[0,T]$ with probability e^{-T}, independently, so almost surely the line splits into finite blocks, each of which evolves on $[0,T]$ as a finite urn. This defines the process for all times, and its law is invariant under translations.

Proposition 8. *For the critical WARM on* $\mathbb{Z}$ *with unit initial counts every count tends to infinity almost surely, and*

$$c := \mathbb{E}\int_0^\infty \frac{dt}{N_{-1}(t)N_0(t)} = \zeta(2) = \frac{\pi^2}{6}.$$

The same expectation, with $N_{v-1}N_v$ *in place of* $N_{-1}N_0$, *equals* $\pi^2/6$ *for every vertex* v *of every even cycle* C_L, $L \geq 4$.

Proof. The proof of Lemma 2 uses only the clocks of finitely many vertices near the edge, so it applies on $\mathbb{Z}$ with $d = 2$ and every count tends to infinity. The count N_0 increases at rate $N_0(S_0^{-1} + S_1^{-1}) \leq 2$, and each increase from n to $n+1$ adds n^{-2} to $A(t) = H^{(2)}(N_0(t) - 1)$, so $\mathbb{E}A(T) = \int_0^T \mathbb{E}\big[(N_0S_0)^{-1} + (N_0S_1)^{-1}\big]dt$. By translation invariance $\mathbb{E}[(N_0S_1)^{-1}] = \mathbb{E}[(N_{-1}S_0)^{-1}]$, and $(N_0^{-1} + N_{-1}^{-1})/S_0 = 1/(N_{-1}N_0)$, so $\mathbb{E}\int_0^T \frac{dt}{N_{-1}N_0} = \mathbb{E}H^{(2)}(N_0(T) - 1)$, which increases to $\zeta(2)$ by bounded convergence. On C_L, $\mathbb{E}\langle M\rangle_\infty = L\zeta(2)$ by Theorem 3, and the L vertex terms have the same expectation by rotation invariance. □

On $\mathbb{Z}$ the kernel is replaced by a local conservation law. With $Y_e = (-1)^e H(N_e - 1)$ and $J_v = (-1)^v / S_v$, the computation in Theorem 1 gives

$$dY_e = -(J_{e+1} - J_e)\, dt + dM_e,$$

with M_e a martingale. Linearising (2) at $x \equiv 1$ gives the decay rate $(1 + \cos k)/2$ for the mode e^{ike}, which vanishes at $k = \pi$. Writing $x_e = 1 + (-1)^e u(e) + w(e)$ with u slowly varying, the field w relaxes to $(u^2)''/8$ and, formally, u obeys

$$\partial_\tau u = \tfrac{1}{4}(1-u^2)\, \partial_e^2 u, \qquad \text{equivalently} \qquad \partial_\tau v = \tfrac{1}{4}\, \partial_e^2 \tanh v, \quad v = \operatorname{artanh} u,$$

which conserves $\sum v$, the continuum form of the charge. Started from the noise of the early stage, this heat flow in $\tau = \log t$ predicts that the local alternating amplitude decays like $(\log t)^{-1/4}$.

5. Numerical checks

All simulations run the process exactly (rate-one vertex clocks, unit initial counts).

Terminal law on cycles. The program `code/exact_law_check.py` runs independent copies of the jump chain of the urn on C_L and compares the charge after K rings with the law of Corollary 7. With $K = 4000$ and 20000 copies on C_4, the variance of the charge is 6.63 (exact $2\pi^2/3 = 6.58$, standard error about 0.07), its excess kurtosis 0.59 (exact $12/(5L) = 0.6$), the Kolmogorov–Smirnov distance to the exact law 0.005 (the 5% level is about 0.010), and $\operatorname{Var} p = 0.0587$ for $p = (1 + \tanh(M/4))/2$ (exact 0.0584). With $K = 10^5$ and 4000 copies on C_{50} the variance is 81.9 (exact 82.2, standard error 1.8) and the Kolmogorov–Smirnov distance 0.008 (the 5% level is about 0.022).

Even cycles in continuous time. The program `code/warm2.c` runs R independent copies to time T (each run stops at the first ring after T) and records $M(T)$, the ring-sampled bracket and $\operatorname{artanh}(u)$. The ring-sampled bracket is the sum over rings of $1/(N_{v-1}N_v)$ for the ringing vertex v before the ring; its mean is $\mathbb{E}\langle M\rangle_T$. The value u is read from $\sum_e (-1)^e \log X_e(T) = L \operatorname{artanh}(u)$.

L	T	R	mean M (s.e.)	$\operatorname{Var} M$	mean bracket	$L\,\mathbb{E}\operatorname{artanh}(u)^2$
100	10^3	400	0.30 (0.64)	161.6	164.3	1.61
400	10^3	200	−2.86 (1.76)	620	657	1.56
100	10^4	200	–	154.9	164.7	–

The terminal values are $\mathbb{E}M_\infty^2 = \mathbb{E}\langle M\rangle_\infty = L\pi^2/6$, that is 164.5 and 658.0, and $L\,\mathbb{E}\operatorname{artanh}(u_\infty)^2 = \pi^2/6 = 1.645$ for every L. The bracket per vertex, 1.64 at both sizes, agrees with $\pi^2/6$; the sample variances have standard errors of about 10% of their size at these R. The difference $\sum_e (-1)^e \log X_e(T) - M(T)$ in (1) has root mean square 0.007 at $T = 10^4$, $L = 100$. At $T = 10^3$ the counts on C_{100} are not yet a pure alternation,

which lowers the last column: the slow modes of (2) near wavenumber π decay like $t^{-\pi^2/L^2}$.

Small graphs. On K_4, where $\dim\ker B = 2$, the standard deviation of each X_e over 2000 runs stays near 0.35 between $T = 10^3$ and $T = 10^5$, while $\max_v |S_v/t - 2|$ falls from 0.25 to 0.08: the limit is a random point of a two-dimensional set of equilibria, as Theorem 3 and Remark 4 describe. On the triangle ($\ker B = 0$) the spread decreases, more slowly than $t^{-1/4}$, towards $(1, 1, 1)$.

The line. On a cycle of length 1001 run to $T = 10^5$ the standard deviation of X_e falls from 0.57 at $t = 12$ to 0.39 at $t = 8.8 \times 10^4$, and $(\log t)^{1/4}$ times it stays between 0.67 and 0.72, rising slowly (a fitted exponent of -0.18 in $\log t$ over this range, which a shift of τ by about 3 or the saturation of tanh would account for); the correlation of neighbouring edges tends to -0.92. These runs are consistent with Kleptsyn's conjecture, with convergence at the logarithmic rate the charges predict.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [7] and working together inside the Brain framework [9], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [8], the Math Brain repository [7] and the Brain framework repository [9].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In November 2026 Claude Opus 5.5

(Anthropic), working in Claude Code, found the charge martingales of Theorem 1, proved the convergence on the even cycle, wrote and ran the simulations of Section 5, and wrote the paper. A Codex agent (OpenAI), refereeing the paper, found the terminal law of Theorem 3 and its proof, the identities for $[M_\sigma]$ and $\operatorname{Var}\langle M_\sigma\rangle_\infty$, the identification $c = \zeta(2)$ in Proposition 8 and the convex proof in Remark 4. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/graph-urn-conserved-charges`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [7].

Appendix A. Notation

Symbol	Meaning	First defined
$G = (V, E)$, d	finite graph, its maximum degree	Section 2
N_e, X_e	count of edge e, normalised count $N_e(t)/t$	Section 2
S_v	vertex sum of counts $\sum_{e\ni v} N_e$	Section 2
B	unsigned vertex–edge incidence matrix	Section 2
$H^{(p)}$, H	$H^{(p)}(n) = \sum_{k=1}^{n} k^{-p}$, harmonic number $H = H^{(1)}$	Section 2
$\sigma \in \ker B$	charge vector	Thm. 1
M_σ, $M_{\sigma,\infty}$	charge martingale, its terminal value	Thms. 1, 3
$\langle M_\sigma\rangle$, $[M_\sigma]$	predictable and optional quadratic variation	Thm. 1
ψ, γ	digamma function, Euler's constant	(1)
$F_w(t)$	number of rings of the clock of w by time t	Lem. 2
$a_e(t)$	internal time of edge e	Lem. 2
D_e	edge martingale $H(N_e - 1) - a_e$	Lem. 2
W_e	independent standard exponential variables	Thm. 3
$P_\theta(n)$	Gamma-ratio martingale	Thm. 3
R	compensated quadratic variation $[M_\sigma] - \langle M_\sigma\rangle$	Thm. 3
P, b, π	fibre $\{x > 0 : Bx = b\}$, its vertex sums, orthogonal projection onto $\ker B$	Rem. 4
C_L	cycle with L edges, L even	Section 3
τ	time $\log t$ of the mean-field equation	(2)
$\mathcal{L}$	Lyapunov function of (2)	(2)
K_e	compensated count of edge e	Section 3
$x^{(u)}$, Σ	equilibrium $x_e^{(u)} = 1 + (-1)^e u$, segment of equilibria	Lem. 5
u_∞, M_∞	limit point on Σ, terminal charge on C_L	Thm. 6
Φ	$\sum_e (-1)^e \log x_e$, the charge as a function of the normalised counts	Thm. 6
Λ_j	independent standard logistic variables	Cor. 7
p	limit share $(1 + u_\infty)/2$ on C_4	Section 3
c	expected bracket per vertex on $\mathbb{Z}$	Prop. 8

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